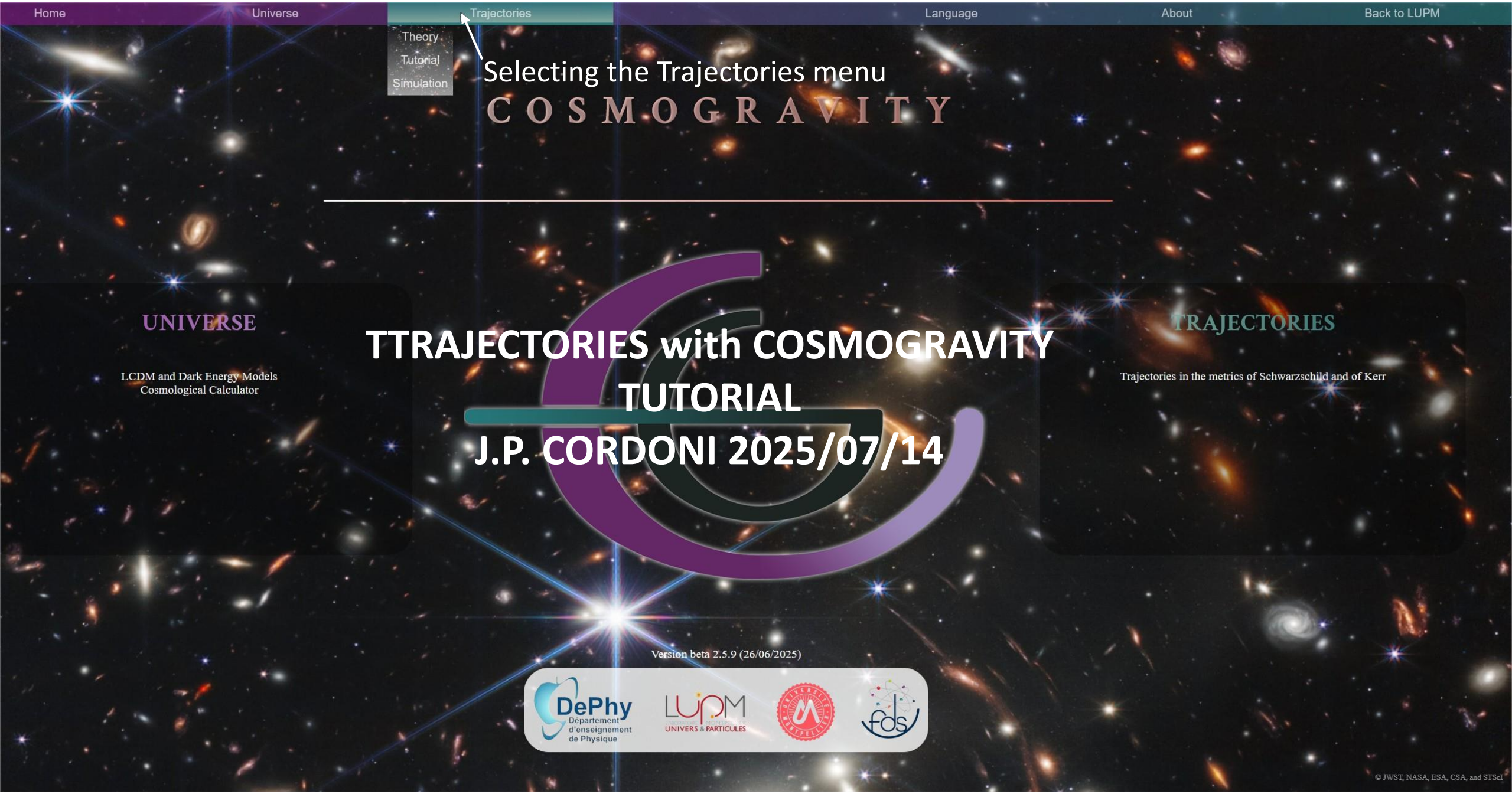
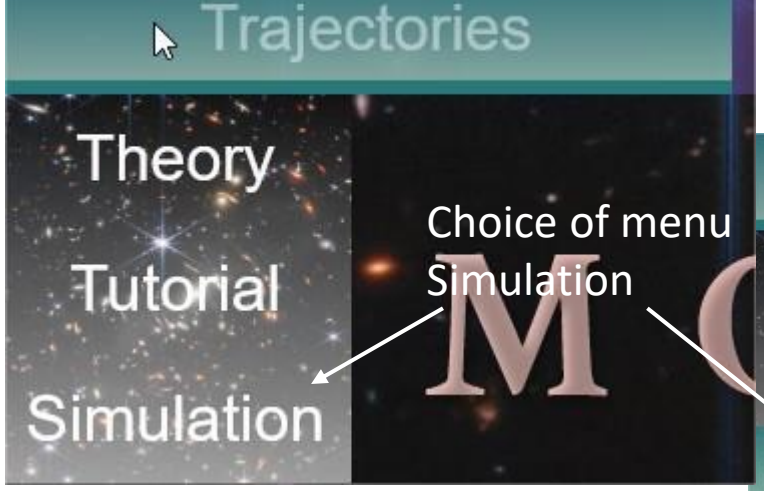
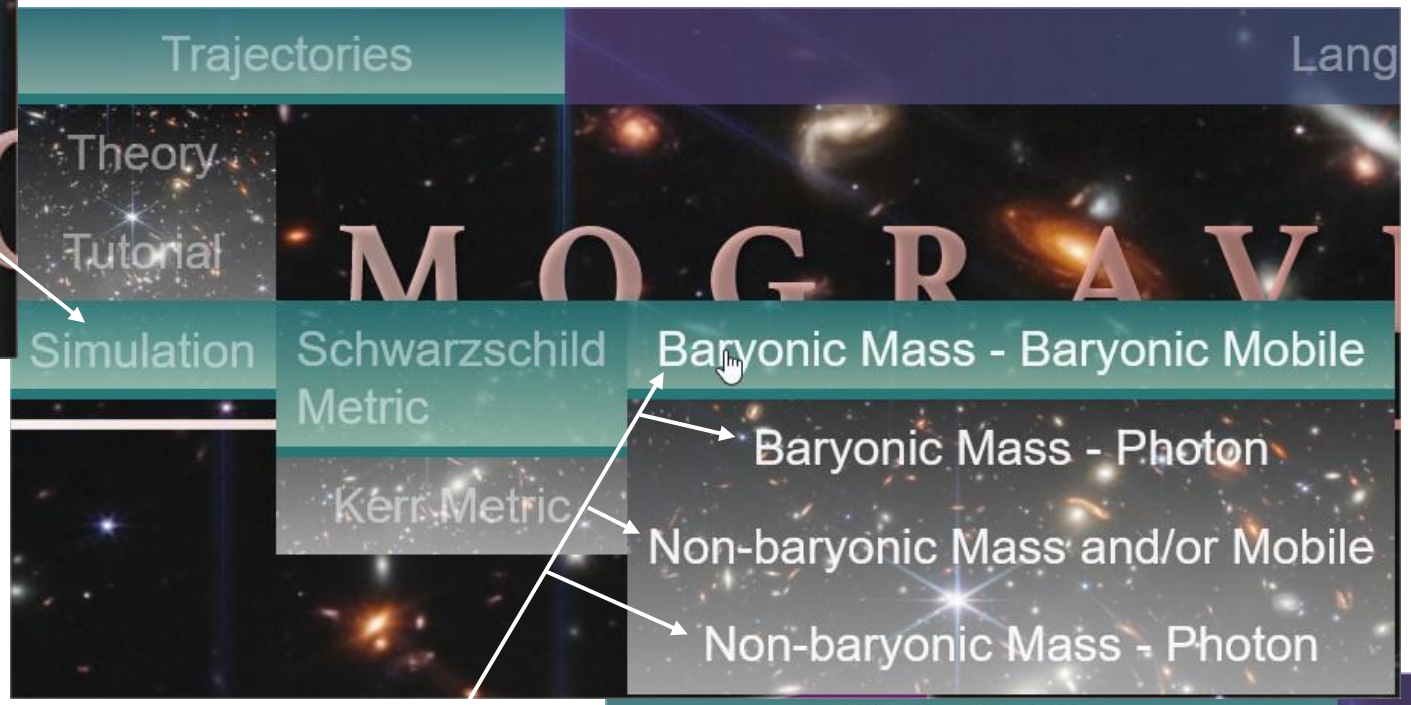


TTRAJECTORIES with COSMOGRAVITY
TUTORIAL
J.P. CORDONI 2025/07/14



Choice of menu Simulation

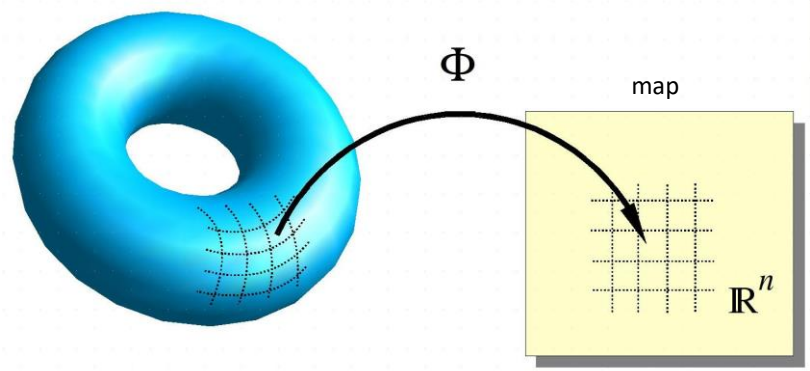


The possibilities for the Schwarzschild metric with the choice of the type of mass and mobile

The possibilities for the Kerr metric with the choice of mobile type

Geometric frame

Relativity has merged space and time, two notions that were completely distinct in Galilean mechanics. Four numbers are needed to determine an event in the space-time continuum: three for its spatial location (e.g. its Cartesian coordinates {x, y, z} or its spherical coordinates {r, θ, φ }) and one for its date (t). The mathematical structure corresponding to this four-dimensional "continuum" is that of **variety**.



Variety: seen closely, a variety looks like R^n (n = 2 on the figure), but this is not necessarily true at the global level.

It should be emphasized that the local similarity with R^4 stops at the labeling of the points and does not extend to the Euclidean space structure of R^4. In particular, the choice of coordinate system is completely free.

These notes are from Gourgoulhon-Relativité Générale

In the **Cosmogravity** software the "trajectories" are the geodesics followed by the different particles (baryonic, non-baryonic, photons) represented by their coordinates (r, φ) in R^2 as a function of the proper time (τ) of the particles or the time of the distant observer (t).
The distance that would be measured (using the scale of the simulation) between two positions of a particle is obviously not equal to the metric distance between these two positions.

Home | Universe | Trajectories | Language | About | Back to LUPM

Central Mass Data

M (kg) = 1.989e30

$r_{physical}$ (m) = 6.96342e6

the Sun

Mobile Inputs

X= 0.003

Display Potential Graph ☐

Mobile 1

r_0 (m) = 1.495978707e11 v_0 (m/s) = 2.97881e+4

Initial distance of projectile to the center of the celestial body

φ_0° = 0 Φ_0° = 90

Use tooltips

The **Save** button saves the graphics and inputs. The **Reset** button ends the simulation and resets the inputs to the default values ... but the **Previous values** button recalls the user's inputs.

Selected simulation

Massive Projectile Trajectory in Schwarzschild Metric

Warning

Read the warning

Full Trajectory | Simple Trajectory | Distant Observer | Spationaut | Rebound

Resume | Reset | Save | Previous values

Continuous trace or point by point

Choose the reference system

Values calculated during simulation

r(m)	Proper time for mobile	Gradient	V_r(m.s^{-1})	V_φ(m.s^{-1})	Distant Observer time	Spectral shift / Energy consumed	$V_{physique}$ (m.s^{-1})	D(m)	g felt	Last g	$P_{consumed}$ (W/kg)
1.496e+11	3.220e+7	3.965e-14	-4.467e+0	2.979e+4	3.220e+7		2.979e+4	9.591e+11	0.000e+0	0.000e+0	0

Calculations paused

Baryonic Mass and Mobile

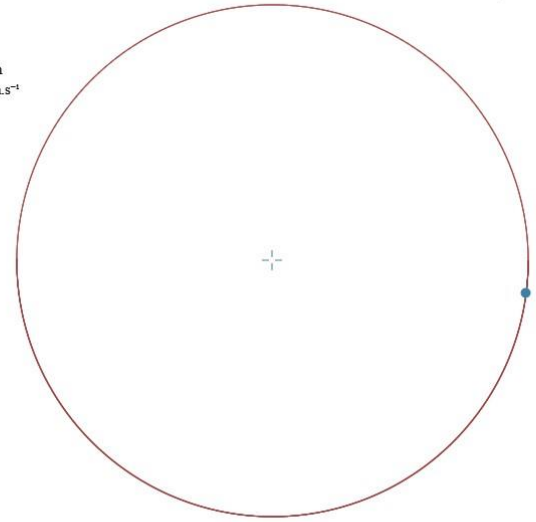
Entries:
M = 1.989e+30 kg
r_{phy} = 6.963e+6 m

Spationaut
mobile1:
r_0 = 1.496e+11 m
V_0 = 2.979e+4 m.s^{-1}
φ_0 = 0.000e+0 °

frame of reference

Scale of simulation

4.0e+10 m



Vφ+

Vφ-

zoom=0

During the simulation you can :

- pause it

- slow it down

simu=1

- speeding it up (which reduces the precision of the trace)

Theoretical Central Mass Data

$rs = \frac{2GM}{c^2}(m)$	$grav = \frac{GM}{R^2}\frac{1}{9.81}(g)$	$Vlib = c(\frac{rs}{R})^{1/2}(m.s^{-1})$
2.954e+3	2.791e+5	6.175e+6

rs: Schwarzschild radius of the Sun

Mobile Outputs

Mobile1		
$L1(m)$	$E1$	$Vcirc1(m/s)$
1.486e+7	1.000e+0	2.97881e+4

For r_0 we have here the speed required to obtain a circular orbit. This value, entered in the v_0 box, is used to plot the orbit : the gravitational field being centro-symmetrical, each circular orbit can be prograde Φ_0=90° or retrograde Φ_0=-90°

Example: The Earth around the Sun

Central Mass Data

M (kg) = 2e13

$r_{physical}$ (m) = 1000

2 mobiles around the asteroid

Mobile Inputs

Number of mobiles 2

Display Potential Graph ☐

Mobile 1

r_0 (m) = 3000 v_0 (m/s) = 0.4

φ_0° = 0 Φ_0° = 65

Mobile 2

r_0 (m) = 5000 v_0 (m/s) = 0.5

φ_0° = 0 Φ_0° = 230

Only the "Distant Observer" choice allows you to have several mobiles.

Selected simulation

Massive Projectile Trajectory in Schwarzschild Metric

Warning

Full Trajectory | Simple Trajectory | Distant Observer | Spationaut | Rebound

Damping coefficient: Limited bounce at impact speed < 300 m/s 0.3

Resume | Reset | Save | Previous values

Bounce with choice of absorption coefficient of impacts

r(m)	Proper time for mobile	Gradient	V_r(m.s^{-1})	V_φ(m.s^{-1})	Distant Observer time	Spectral shift / Energy consumed	$V_{physique}$ (m.s^{-1})
9.998e+2	1.877e+4	2.597e-6	-1.420e-1	1.530e-1	1.876e+4	0.000e+0	2.087e-1
r(m)	Proper time for mobile	Gradient	V_r(m.s^{-1})	V_φ(m.s^{-1})	Distant Observer time	Spectral shift / Energy consumed	$V_{physique}$ (m.s^{-1})
5.244e+3	1.892e+4	3.962e-9	3.029e-1	-3.652e-1	1.892e+4	0.000e+0	4.745e-1

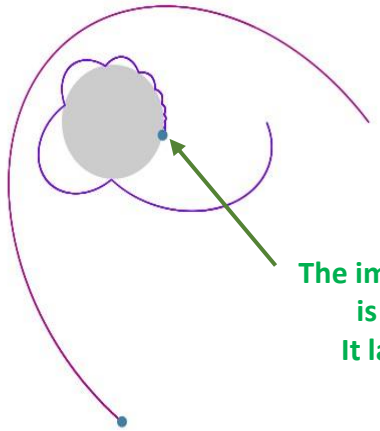
Calculations paused

Baryonic Mass and Mobile

Entries:
M = 2.000e+13 kg
r_{phy} = 1.000e+3 m
Damping coefficient: Limited bounce at impact speed < 300 m/s = 0.3
Distant Observer
mobile1:
r_0 = 3.000e+3 m
V_0 = 4.000e-1 m.s^{-1}
φ_0 = 0.000e+0 °
mobile2:
r_0 = 5.000e+3 m
V_0 = 5.000e-1 m.s^{-1}
φ_0= 0.000e+0°

frame of reference

1.0e+3 m



The impact speed of mobile 1 is less than 300 m/s : It lands on the asteroid

Zoom : erases the previous trajectory

zoom=0 simu=-1

Theoretical Central Mass Data

$rs = \frac{2GM}{c^2}(m)$	$grav = \frac{GM}{R^2}\frac{1}{9,81}(g)$	$Vlib = c(\frac{rs}{R})^{1/2}(m.s^{-1})$
2.970e-14	1.361e-4	1.634e+0

Mobile Outputs

Mobile1		
$L1(m)$	$E1$	$Vcirc1(m/s)$
3.628e-6	1.000e+0	6.67026e-1
Mobile2		
$L2(m)$	$E2$	$Vcirc2(m/s)$
-6.388e-6	1.000e+0	5.16676e-1

The E and L constants are calculated as a function of r_0 , rs , a , v_0 , Φ_0 (see theory)

Example: Small asteroid

Central Mass Data

M (kg) = 2e30

$r_{physical}$ (m) = 0

A mass of zero physical radius is a black hole

Mobile Inputs

Number of mobiles 2

Display Potential Graph

Photon 1

r_0 (m) = 10000 φ_0° = 120

$\dot{\varphi}_0$° = -135

Photon 2

r_0 (m) = 10000 φ_0° = 135

$\dot{\varphi}_0$° = 210

Selected simulation

Photon Trajectory in Schwarzschild Metric

Warning

Full Trajectory | Simple Trajectory | Distant Observer | Photon

Resume | Reset | Save | Previous values

r(m)	Proper time for mobile	V_r(m.s^{-1})	V_φ(m.s^{-1})	Distant Observer time	$V_{physique}$ (m.s^{-1})
1.176e+4	0.000e+0	2.352e+8	-1.859e+8	1.276e-4	2.99792458e+8
r(m)	Proper time for mobile	V_r(m.s^{-1})	V_φ(m.s^{-1})	Distant Observer time	$V_{physique}$ (m.s^{-1})
2.970e+3	0.000e+0	2.998e+8	0.000e+0	1.276e-4	2.99792458e+8

The proper time of a photon is always zero

The speed of the photon is constant and identical in all reference frames

Calculations paused

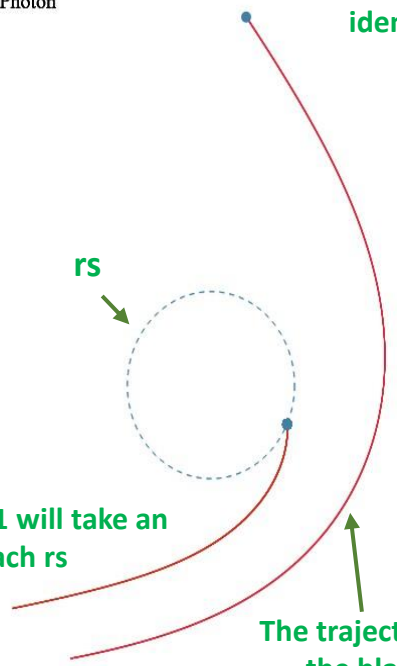
Baryonic Mass and Photon

Entries:
M = 2.000e+30 kg
r_{phy} = 0.000e+0 m

frame of reference

Distant Observer
mobile1:
r_0 = 1.000e+4 m
V_0 = 2.998e+8 m.s^{-1}
φ_0 = 1.200e+2 °
mobile2:
r_0 = 1.000e+4 m
V_0 = 2.998e+8 m.s^{-1}
φ_0= 1.350e+2 °

3.0e+3 m



For the observer, photon 1 will take an infinite time to reach rs

The trajectory of photon 2 is deflected in the black hole's gravitational field

zoom=0 simu=0

Theoretical Central Mass Data

$rs = \frac{2GM}{c^2}(m)$	$T = 6.15 * 10^{-8} \frac{M_\odot}{M}(K)$	$t = 6.6 * 10^{74} (\frac{M}{M_\odot})^3 (s)$
2.970e+3	6.464e-8	6.710e+74

Mobile Outputs

Photon1		
$L1(m)$	$E1$	$rc1(m)$
-8.434e+3	1.000e+0	4.45540e+3
Photon2		
$L2(m)$	$E2$	$rc2(m)$
-5.963e+3	1.000e+0	4.45540e+3

Radius of possible circular orbit
prograde Φ_0=90°
or
retrograde Φ_0=-90°

Example: Photon trajectories

Central Mass Data

M (kg) = 2e30

$r_{physical}$ (m) = 7e8

The physical radius is greater than rs

Mobile Inputs

Number of mobiles: 3

Display Potential Graph ☐

Mobile 1

r_0 (m) =	9e8	v_0 (m/s)=	3e5
φ_0° =	0	Φ_0° =	90

Mobile 2

r_0 (m) =	4e8	v_0 (m/s)=	3e5
φ_0° =	90	Φ_0° =	90

Mobile 3

r_0 (m) =	6e8	v_0 (m/s)=	3e5
φ_0° =	180	Φ_0° =	90

The three particles of baryonic matter are subject only to the gravitational field of the central mass

Selected simulation

Massive Projectile Trajectory in Schwarzschild Metric
(non-baryonic case)

Warning

Full Trajectory | Simple Trajectory | Distant Observer | Spationaut

Resume | Reset | Save | Previous values

r(m)	Proper time for mobile	Gradient	V_r(m.s^{-1})	V_φ(m.s^{-1})	Distant Observer time	Spectral shift / Energy consumed	$V_{physique}$ (m.s^{-1})
8.936e+8	2.250e+4	3.114e-8	2.872e+4	3.021e+5	2.250e+4	2.174e-6	3.035e+5
r(m)	Proper time for mobile	Gradient	V_r(m.s^{-1})	V_φ(m.s^{-1})	Distant Observer time	Spectral shift / Energy consumed	$V_{physique}$ (m.s^{-1})
4.801e+8	2.250e+4	1.202e-6	9.446e+3	2.499e+5	2.250e+4	3.031e-6	2.501e+5
r(m)	Proper time for mobile	Gradient	V_r(m.s^{-1})	V_φ(m.s^{-1})	Distant Observer time	Spectral shift / Energy consumed	$V_{physique}$ (m.s^{-1})
4.823e+8	2.250e+4	2.185e-6	-1.698e+4	3.732e+5	2.250e+4	3.455e-6	3.736e+5

Calculations paused

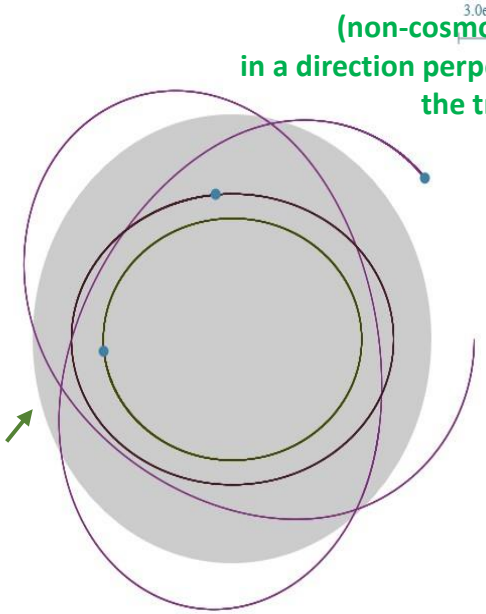
To calculate the spectral shift, the observer is assumed to be very far away (non-cosmological distance) in a direction perpendicular to the plane of the trajectories

Mass and/or non-baryonic mobile(s)

Entries:
M = 2.000e+30 kg
r_{phy} = 7.000e+8 m
Distant Observer
mobile1:
r_0 = 9.000e+8 m
V_0 = 3.000e+5 m.s^{-1}
φ_0 = 0.000e+0 °
mobile2:
r_0 = 4.000e+8 m

V_0 = 3.000e+5 m.s^{-1}
φ_0= 9.000e+1 °

frame of reference



physical radius

zoom=0

simu=-3

Theoretical Central Mass Data

$rs = \frac{2GM}{c^2}(m)$	$grav = \frac{GM}{R^2}\frac{1}{9.81}(g)$	$Vlib = c(\frac{rs}{R})^{1/2}(m.s^{-1})$
2.970e+3	2.777e+1	6.175e+5

Mobile Outputs

Mobile1		
$L1(m)$	$E1$	$Vcirc1(m/s)$
9.006e+5	1.000e+0	3.85108e+5
Mobile2		
$L2(m)$	$E2$	$Vcirc2(m/s)$
4.003e+5	1.000e+0	2.49527e+5
Mobile3		
$L3(m)$	$E3$	$Vcirc3(m/s)$
6.004e+5	1.000e+0	3.74290e+5

**Example:
Matter of the Sun assumed non-baryonic of constant mass density**

Home | Universe | Trajectories | Language | About | Back to LUPM

Central Mass Data

M (kg) = 1.9e38

J (kg.m^2.s^{-1}) = 7.8e57

Mobile Inputs

X= 0.003

Display Potential Graph

r_0 (m) = 5e11

V (m.s^{-1}) = 1.84e8

φ_0°= 0

$\dot{\varphi}_0$° = 90

The particle can emerge from the ergoregion, but remains trapped in the black hole's gravitational field

Selected simulation

Massive Particle Trajectory in Kerr Metric

Warning

Full Trajectory | Simple Trajectory | Distant Observer | Spationaut | Exceed Rh+

Resume | Reset | Save | Previous values

r(m)	Proper time for mobile	Acceleration gradient	V_r(m.s^{-1})	V_φ(m.s^{-1})	Distant Observer time	Spectral shift / Energy consumed	$V_{physique}$(m.s^{-1})	D(m)	g felt	Last g	$P_{consumed}$ (W/kg)
4.586e+11	1.769e+4	3.686e-8	2.764e+7	1.991e+8	4.520e+4		2.010e+8	4.476e+12	0.000e+0	0.000e+0	0

Calculations paused

Massive Particle Trajectory in Kerr Metric

Entries:
M = 1.900e+38 kg
J = 7.800e+57 kg.m².s⁻¹
a = 1.369e+11 m
r_0 = 5.000e+11 m
V_0 = 1.840e+8 m.s⁻¹
φ_0 = 0.000e+0 °
Spationaut

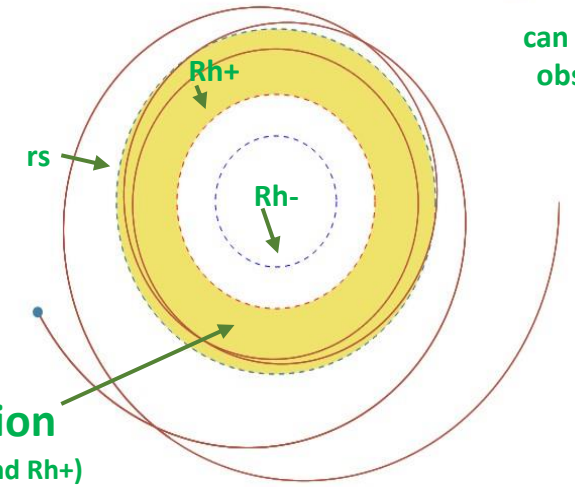
Note the large time lag between the particle's proper time and the observer's time

frame of reference

1.0e+11 m

rs
Rh+
Rh -

The particle's velocity can be measured by a static observer only outside the radius rs



Ergorégion
(between rs and Rh+)

Vφ+

Vφ-

zoom=0

simu=-1

Theoretical Central Mass Data

$rs = \frac{2GM}{c^2}(m)$	$Rh+(m)$	$Rh-(m)$
2.822e+11	1.751e+11	1.071e+11
$a = \frac{J}{cM}(m)$	$g = \frac{1}{2}c^2\frac{(Rh+^3-a^2)}{(Rh+^3+a^2)Rh+}(m.s^{-2})$	
1.369e+11	6.181e+4	

Mobile Outputs

$L(m)$	E
2.725e+11	8.360e-1
$V_{prograde}(m)$	
1.95826e+8	

This is the speed on the circular orbit of radius r_0

Example: Massive particle and spinning black hole

Home | Universe | Trajectories | Language | About | Back to LUPM

Central Mass Data

M (kg) = 2e30

$r_{physical}$ (m) = 0

A mass of zero physical radius is a black hole

Mobile Inputs

Number of mobiles 2

Display Potential Graph ☑

Choose an option: Potential 1

Allows real-time selection of the potential graph to display

Mobile 1

r_0 (m) = 6000 v_0 (m/s) = 2.09894e+8

φ_0° = 0 Φ_0° = 90

Mobile 2

r_0 (m) = 10000 v_0 (m/s) = 1.37795e+8

φ_0° = 0 Φ_0° = 90

We choose the radii r_0 for each mobile and then transfer the velocities calculated in "Mobile Outputs" to the v_0 boxes to obtain two circular orbits.

The graph of the first mobile's potential shows that its trajectory is unstable.
In real time, the choice of the second mobile's potential allows us to "see" that its trajectory is stable.

Selected simulation

Massive Projectile Trajectory in Schwarzschild Metric

Warning

Full Trajectory | Simple Trajectory | Distant Observer | Spationaut

Resume | Reset | Save | Previous values

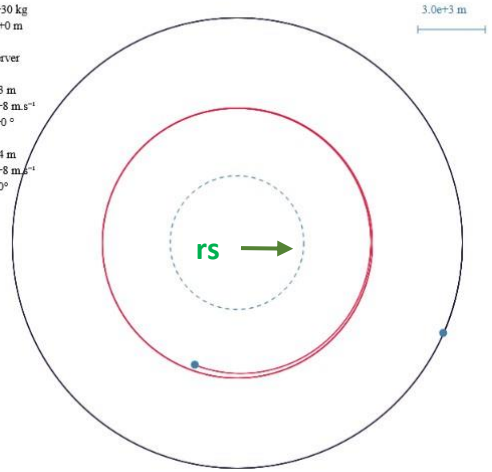
r(m)	Proper time for mobile	Gradient	V_r(m.s^{-1})	V_φ(m.s^{-1})	Distant Observer time	Spectral shift / Energy consumed	$V_{physique}$ (m.s^{-1})
5.736e+3	2.928e-4	3.373e+8	-9.701e+6	2.145e+8	5.795e-4	1.064e+0	2.148e+8
r(m)	Proper time for mobile	Gradient	V_r(m.s^{-1})	V_φ(m.s^{-1})	Distant Observer time	Spectral shift / Energy consumed	$V_{physique}$ (m.s^{-1})
1.000e+4	4.315e-4	1.454e+7	-3.188e+2	1.378e+8	5.795e-4	3.430e-1	1.378e+8

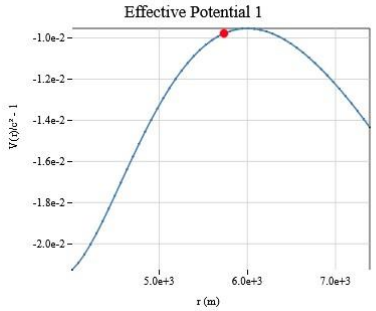
Calculations paused

Baryonic Mass and Mobile

Entries:
M = 2.000e+30 kg
r_{phy} = 0.000e+0 m

Distant Observer
mobile1:
r_0 = 6.000e+3 m
V_0 = 2.099e+8 m.s^{-1}
φ_0 = 0.000e+0 °
mobile2:
r_0 = 1.000e+4 m
V_0 = 1.378e+8 m.s^{-1}
φ_0= 0.000e+0°





zoom=0 simu=0

Theoretical Central Mass Data

$rs = \frac{2GM}{c^2}(m)$	$T = 6.15 * 10^{-8} \frac{M_\odot}{M}(K)$	$t = 6.6 * 10^{74} (\frac{M}{M_\odot})^3 (s)$
2.970e+3	6.464e-8	6.710e+74

Mobile Outputs

Mobile1		
$L1(m)$	$E1$	$Vcirc1(m/s)$
5.883e+3	9.952e-1	2.09894e+8
Mobile2		
$L2(m)$	$E2$	$Vcirc2(m/s)$
5.175e+3	9.441e-1	1.37795e+8

These are the speeds on circular orbits for two chosen radii
prograde Φ_0=90°
ou
retrograde Φ_0=-90°

Example: Simulation of two circular trajectories

Home | Universe | Trajectories | Language | About | Back to LUPM

Central Mass Data

M (kg) = 5.97e2

$r_{physical}$ (m) = 6378e3

The Earth

Mobile Inputs

The satellite

X= 0.001

Display Potential Graph

Mobile 1

r_0 (m) = 6878e3 v_0 (m/s) = 7.61105e+3

φ_0° = 0 $\dot{\varphi}_0$° = 90

Selected simulation

Massive Projectile Trajectory in Schwarzschild Metric

Warning

Full Trajectory | Simple Trajectory | Distant Observer | Spationaut | Rebound

Resume | Reset | Save | Previous values

r(m)	Proper time for mobile	Gradient	$V_r(m.s^{-1})$	$V_\varphi(m.s^{-1})$	Distant Observer time	Spectral shift / Energy consumed	$V_{physique}(m.s^{-1})$	D(m)	g felt	Last g	$P_{consumed}$ (W/kg)
9.437e+6	1.655e+4	2.513e-7	2.527e+1	5.967e+3	1.655e+4	5.059e-11	5.967e+3	1.188e+8	0.000e+0	1.669e+1	1.338e+6

Calculations paused

Baryonic Mass and Mobile

Entries:
M = 5.970e+24 kg
r_{phy} = 6.378e+6 m

Spationaut
mobile1:
r_0 = 6.878e+6 m
V_0 = 7.611e+3 m.s^{-1}
φ_0 = 0.000e+0 °

2.0e+6 m

Energy expended during the piloting time
5,059^e-11 * c^2 * m
= 0,455^{e}10 J

Elliptical orbit
2a=16315^{e}3 m
Measured by r(m)

Initial circular orbit
500 km from the earth's surface
2a=13756^{e}3 m

When the satellite has completed one revolution, the tangential velocity is increased,
at a rate defined by the variable X,
by several clicks
on the upper joystick
(Vphi+).

Joysticks
(for spationaut only)

Increases tangential speed → Vφ+

Vφ- ← Reduces tangential speed

zoom=0

simu=-4

Theoretical Central Mass Data

$rs = \frac{2GM}{c^2}(m)$	$grav = \frac{GM}{R^2}\frac{1}{9.81}(g)$	$Vlib = c(\frac{rs}{R})^{1/2}(m.s^{-1})$
8.866e-3	9.984e-1	1.118e+4

Mobile Outputs

Mobile1		
$L1(m)$	$E1$	$Vcirc1(m/s)$
1.878e+2	1.000e+0	7.61105e+3

Speed
on initial circular orbit

Energy on initial circular orbit
- 2,896^{e}10 J

Energy on the elliptical orbit
- 2,442^{e}10 J

Energy difference
+ 0,454^{e}10 J

In Newtonian gravitation (which is the case here), the energy of a moving body of mass m on an elliptical trajectory with axis 2a is equal to:
- G M m / 2a

For example, let's take m=1000 kg

Example:
Piloting a satellite From circular to elliptical orbit

Home | Universe | Trajectories | Language | About | Back to LUPM

Central Mass Data

M (kg) = 8e36

$r_{physical}$ (m) = 0

A mass of zero physical radius is a black hole

Mobile Inputs

Display Potential Graph ☐

Photon 1

r_0 (m) = 6e10 φ_0° = -28.275

Φ_0° = 208.275

Initial conditions chosen so that the paths "outward" and "return" are "parallel".

Selected simulation

Photon Trajectory in Schwarzschild Metric

Warning

Full Trajectory | Simple Trajectory | Distant Observer | Photon

Resume | Reset | Save | Previous values

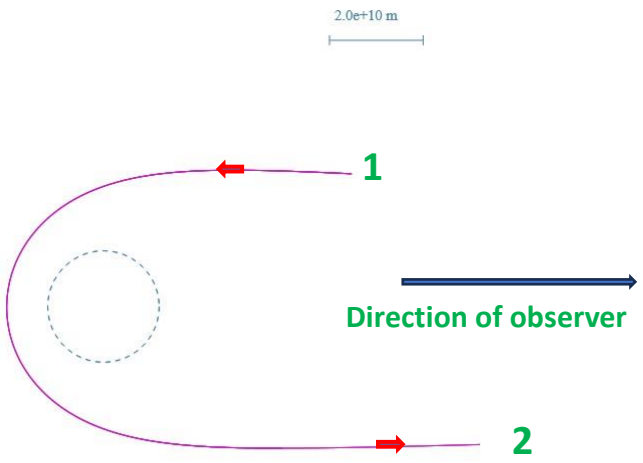
r(m)	Proper time for mobile	V_r(m.s^{-1})	V_φ(m.s^{-1})	Distant Observer time	$V_{physique}$ (m.s^{-1})
9.530e+10	0.000e+0	2.849e+8	-9.341e+7	6.080e+2	2.99792458e+8

Calculations paused

Baryonic Mass and Photon

Entries:
M = 8.000e+36 kg
r_{phy} = 0.000e+0 m

Photon
mobile1:
r_0 = 6.000e+10 m
V_0 = 2.998e+8 m.s^{-1}
φ_0 = -2.827e+1 °



The photon coming from the observer (1), deflected by the black hole's intense gravitational field, returns in the direction of the observer (2).

zoom=0 simu=1

Theoretical Central Mass Data

$rs = \frac{2GM}{c^2}(m)$	$T = 6.15 * 10^{-8}\frac{M_\odot}{M}(K)$	$t = 6.6 * 10^{74}(\frac{M}{M_\odot})^3(s)$
1.188e+10	1.616e-14	4.294e+94

Mobile Outputs

Photon1		
$L1(m)$	$E1$	$rc1(m)$
-3.174e+10	1.000e+0	1.78216e+10

Circular orbit radius
prograde Φ_0=90°
or
retrograde Φ_0=-90°

Trajectory "round trip" of a photon

Home | Universe | Trajectories | Language | About | Back to LUPM

Central Mass Data

M (kg) = 8e36

J (kg.m^2.s^{-1}) = 1.4e55

Mobile Inputs

Display Potential Graph ☐

r_0 (m) = 5e13

φ_0°= 0.018

$\dot{\varphi}_0$°= 179.98

The photon coming from the observer, deflected by the black hole's very intense angular momentum, will never return towards the observer.

Selected simulation

Photon Trajectory in Kerr Metric

Warning

Full Trajectory | Simple Trajectory | Distant Observer | Photon | Exceed Rh+ ☐

Resume | Reset | Save | Previous values

r(m)	Proper time for mobile	V_r(m.s^{-1})	V_φ(m.s^{-1})	Distant Observer time	$V_{physique}$(m.s^{-1})
1.994e+10	0.000e+0	2.129e+8	2.111e+8	1.669e+5	2.99792458e+8

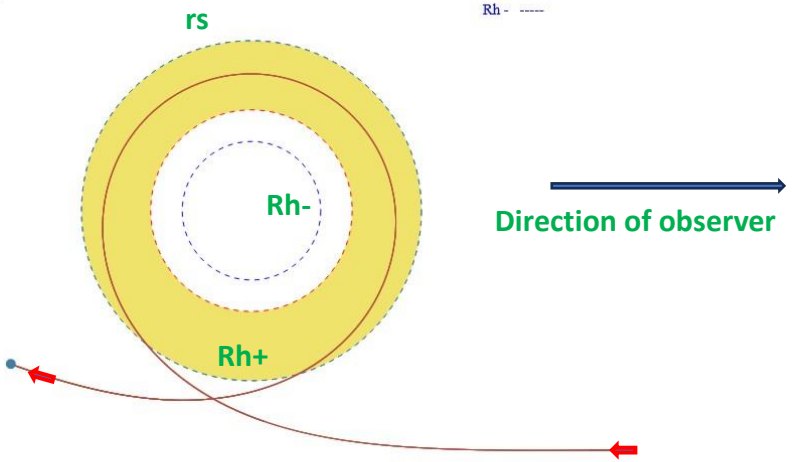
Calculations paused

Photon Trajectory in Kerr Metric

Entries:
M = 8.000e+36 kg
J = 1.400e+55 kg.m².s⁻¹
a = 5.837e+9 m
r_0 = 5.000e+13 m
V_0 = 2.998e+8 m.s⁻¹
φ_0 = 1.800e-2 °
Photon

6.0e+9 m

rs -----
Rh+ -----
Rh - -----



The photon can emerge from the ergoregion

Direction of observer

Initially the photon is outside the simulation window : we have zoomed in sharply when the photon becomes close to rs

zoom=43

simu=-12

Theoretical Central Mass Data

$rs = \frac{2GM}{c^2}(m)$	$Rh+(m)$	$Rh-(m)$
1.188e+10	7.043e+9	4.838e+9
$a = \frac{J}{cM}(m)$	$g = \frac{1}{2}c^2\frac{(Rh+^2-a^2)}{(Rh+^2+a^2)Rh+}(m.s^{-2})$	
5.837e+9	1.184e+6	

Mobile Outputs

$L(m)$	E
1.571e+10	1.000e+0
$r_{prograde}(m)$	$r_{retrograde}(m)$
7.26334e+9	2.36703e+10

Circular orbit radius
prograde Φ_0=90°
or
retrograde Φ_0=-90°

Trajectory "outward" without "return" of a photon

Home | Universe | Trajectories | Language | About | Back to LUPM

Central Mass Data

M (kg) = 2e39

J (kg.m².s⁻¹) = 8.90462694e59

Mobile Inputs

X= 0.003

Display Potential Graph ☑

r_0 (m) = 1.48519969e12

V (m.s⁻¹) = 1.75e8

φ_0°= 0

$\dot{\varphi}_0$° = 90

For Cosmogravity,
the angular momentum limit is:
$J_{limite} = c\,M\,rs / 2$
(when a = rs / 2 : see theory)

Here we choose :
an angular momentum J
such that :
$J_{limite} - J = 1^{e}\text{-}8$

and a starting radius
such that :
$r_0 - Rh+ = 1^{e}\text{-}8$

Massive Particle Trajectory in Kerr Metric

Warning

Full Trajectory | Simple Trajectory | Distant Observer | Spationaut

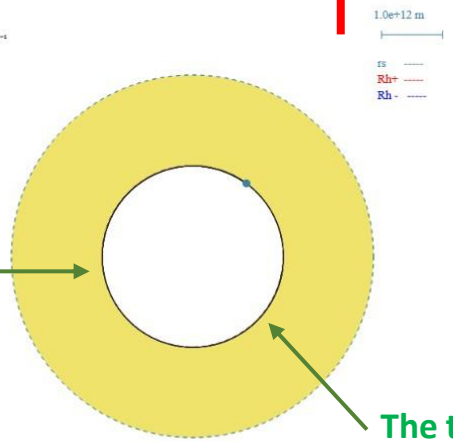
Resume | Reset | Save | Previous values

r(m)	Proper time for mobile	Acceleration gradient	V_r(m.s⁻¹)	V_φ(m.s⁻¹)	Distant Observer time	Spectral shift / Energy consumed	$V_{physique}$(m.s⁻¹)
1.485e+12	1.000e+0	1.765e-16	undefined	undefined	3.539e+6	Infinity	undefined

Calculations paused

Massive Particle Trajectory in Kerr Metric

Entries:
M = 2.000e+39 kg
J = 8.905e+59 kg.m².s⁻¹
a = 1.485e+12 m
r_0 = 1.485e+12 m
V_0 = 1.750e+8 m.s⁻¹
φ_0 = 0.000e+0 °
Distant Observer

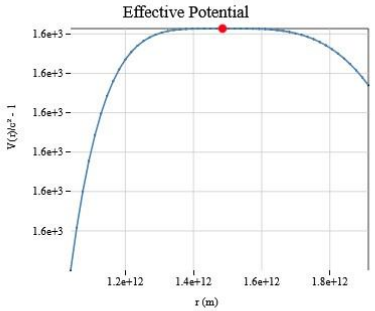


Rh+ and Rh- are indistinguishable

The observer's time has varied by about 40 days for one second of the spationaut's proper time.

The trajectory is at the "limit" of Rh+

Trajectory « very stable »



zoom=-5

simu=0

Theoretical Central Mass Data

$rs = \frac{2GM}{c^2}(m)$	$Rh+(m)$	$Rh-(m)$
2.970e+12	1.485e+12	1.485e+12
$a = \frac{J}{cM}(m)$	$g = \frac{1}{2}c^2\frac{(Rh+^2-a^2)}{(Rh+^2+a^2)Rh+}(m.s^{-2})$	
1.485e+12	1.196e+0	

Mobile Outputs

$L(m)$	E
1.199e+14	4.038e+1

If the starting point
of the massive particle
is between rs and Rh+ ,
only circular trajectories are simulated.
The values of E and L are calculated
as a function of
r_0, rs , a
(see theory)

Trajectory at « limit »